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Samoa Food Away From Home (FAFH)

PACIFIC COMMUNITY

INNOVATIVE EXPERIMENT REPORT



Samoa Food Away From Home (FAFH) – Innovative Experiment report

Nathalie Troubat¹, Michael Sharp², Lilianetelani Hennemann³



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¹ Harmonisation Advisor, Statistics for Development Division, Pacific Community (nathalietr@spc.int)

² Statistics Adviser (Economics), Statistics for Development Division, Pacific Community (michaels@spc.int)

³ Assistant Chief Executive Officer, Economic Statistics Division, Samoa Bureau of Statistics
(lilianetelani.hennemann@sbs.gov.ws)

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Acronyms

AFH	Away from home consumption
CAPI	Computer assisted personal interview
CV	Coefficient of variation
DEC	Dietary energy consumption
DEC_AFH	Dietary energy consumed away from home
FAFH	Foods prepared and consumed away from home, or food away from home
FAO	Food and Agriculture Organization of the United Nations
HCES	Household Consumption and Expenditure Survey
HH	Household
HIES	Household Income and Expenditure Survey
MDER	Minimum dietary energy requirements
PacStat	Statistical Innovation and Capacity Building in the Pacific Islands Project
PAPI	Paper and pencil interviewing
PICTs	Pacific Islands countries and territories
PoU	Prevalence of undernourishment
RMI	Republic of Marshall Islands
SBS	Samoa Bureau of Statistics
SDG	Sustainable Development Goals
WB	World Bank

Executive summary

SPC in collaboration with the Samoa Bureau of Statistics conducted an experiment in Samoa on food away from home consumption using innovative approaches such as web search and crowd sourcing.

The aim of the experiment was twofold: i. to improve the current methodology used to estimate the dietary energy (calories) consumed away from home using information collected in a household income and expenditure survey (HIES); and ii. to propose an innovative protocol to better estimate away from home (AFH) food consumption.

Through the experiment, we found that, relative to at home food consumption, fewer calories per amount spent are sourced from AFH consumption, and less than had previously been assumed. In fact, we found that the cost of one calorie consumed away from home was more than twice higher than one calorie consumed at home (after controlling for socio-economic characteristics of the household), which is far greater than the 1.25 multiplier currently assumed and used when processing food data collected in HIES conducted in the Pacific Island countries and territories (PICTs).

This means that existing methods of estimating calories consumed using HIES data tend to over-estimate the AFH dietary energy consumption (DEC), which leads to under estimation of the calorie deficits in a population.

Through this experiment we therefore learned that:

- A multiplier or adjustment factor is needed when using an indirect approach (i.e., the status quo in food data processing in the PICTs) to estimate the DEC AFH from data collected in HIES.
- The multiplier estimated through the experiment is higher than the current multiplier used in the analysis of recent HIES conducted in the PICTs and the implication of the choice of the multiplier matters with respect to estimation of the cost and magnitude of DEC which in turn affect estimates of the Prevalence of Undernourishment (PoU) and the prevalence of monetary-based poverty using the cost of basic needs approach. Which multiplier to use when analysing food data collected in HIES remains an unanswered question that can be addressed only through the implementation of similar experiment in other PICTs.
- It is possible to collect information on foods prepared and consumed away from home in a HIES using a direct approach through the inclusion of a separate module combining an individual diary and visual aid. Such approach, despite the challenges, has been successfully implemented in Samoa and could be deployed in other PICTs.

It is recommended that, for future HIES conducted in the PICTs, a higher multiplier is applied than is currently used when estimating the dietary energy consumed away from home. Depending on the context of AFH consumption, the results of the Samoa experiment suggest that a multiplier close to 2 might be more appropriate.

1. Introduction

‘Food Away From Home’ (FAFH) refers to food and beverage that is acquired and consumed away from the dwelling, such as at a restaurant or café, the home of a friend or family member, and at feasting events, such as work, Church, or a wedding. FAFH is differentiated from ‘in-house’ consumption, with the statistical unit being the individual, whereas consumption at home is measured at the household level. Anecdotal evidence suggests that FAFH is an increasingly significant source of dietary energy among Pacific Islanders. However current FAFH data collection and processing methodologies, and assumptions used, are potentially introducing errors in the estimation of dietary energy sourced from FAFH consumption. This has adverse implications for measuring poverty (SDG 1.1 and 1.2) and undernourishment (SDG 2.1.1), and the issue remains unresolved by the Pacific Statistics Methods Board (ref: Himelein, K, (2020)). This Innovative Experiment aims to test the assumptions used in the estimation of DEC sourced from FAFH consumption to improve the measurement of total DEC (i.e., both in-house plus FAFH food consumption) in the Pacific Island countries and territories (PICTs) and based on data sourced from Household Income and Expenditure Surveys (HIES).

With the development of modern economies, consuming food and beverages outside of the household’s dwelling, beyond traditional cultural practices, has become common consumer behaviour in the Pacific, contributing up to 15% of the total amount of dietary energy consumption in some countries like Marshall Islands (RMI) or Wallis and Futuna. However, the lack of adequate tools to collect information on foods prepared and consumed away from home (FAFH), including beverages, and the absence of a proper method to estimate dietary energy from these foods poses significant challenges for data providers and analysts.

So far, most Household Consumption and Expenditure Surveys (HCES) conducted worldwide have collected information on FAFH through various approaches that were considered “ad hoc and unsatisfactory”, as evidenced by Smith and others (2014), who assessed food data collected from 100 HCES. They found that, although expenditure on FAFH could be estimated for 90 percent of the surveys, metric quantities could only be estimated for less than 10 percent of the surveys assessed.

Quantities of FAFH consumption are needed to estimate total dietary energy consumption, which, in turn, is required to analyse food consumption patterns to measure poverty and estimate the prevalence of undernourishment. In the absence of data on individual food consumption, poverty, and food security indicators are informed by the food data collected in HCES. However, their measurement, will be biased if food data is not appropriately collected. Any improvement in the way food data is collected in HCES positively impacts the accuracy of the estimation of malnutrition and poverty.

To address this issue, the Food and Agriculture Organization of the United Nations (FAO) and the World Bank (WB) drafted recommendations to improve the design of HCES surveys to collect better quality food data (hereafter referred to as the WB/FAO guidelines (FAO and WB, 2018)⁴). The guidelines recommend that FAFH be captured in a separate survey module, asking each household member to report on the number of meals consumed away from home by meal event (i.e., breakfast, lunch, dinner, snacks, etc.) and the respective amount spent. This module was then developed to help respondents remember and better report on their consumption outside the dwelling, thus overcoming the issue of under-reporting observed in pre-existing methods used to capture FAFH (Farfan et al., 2019). The module was also designed to be easy to administer. At the time the recommendations were formed,

⁴ In 2018, the UN Statistical Commission endorsed data collection guidelines to better capture food consumption in HCES and to improve the quality of statistics used to inform poverty, food security, and nutrition analyses, while maintaining the traditional purpose to derive the weights for use in the rebase of consumer price indices (CPIs) and in the compilation of national accounts. The WB/FAO guidelines, besides providing recommendations on how to better capture food consumption data, aim to harmonize survey design worldwide to derive indicators that can be compared between countries, and over time. The guidelines also bring up the issue of harmonizing the processing of food data resulting from HCES. The guidelines can be downloaded at: <https://www.worldbank.org/en/programs/lsm/publication/Food-data-collection-in-household-consumption-and-expenditure-surveys-guidelines-for-low-and-middle-income-countries>

the benefit of adding such a module in the HCES was not yet demonstrated, and the guidelines concluded that more research work was needed to ensure that data collected on FAFH provides an accurate measure of its contribution to estimating overall consumption.

From June 2018 to April 2019, the Pacific Community (SPC), in collaboration with the World Bank, conducted a Household Income and Expenditure Survey (HIES)⁵ experiment in RMI to test optimized methods for collecting socio-economic data following the WB/FAO guidelines. In the RMI experiment, food away from home was captured through a separate module, asking each household member to recall the number of meals consumed away from home over the last 7 days by meal event, and the respective amount of money spent, as recommended in the WB/FAO guidelines (hereafter referred to as “AFH - 7 day recall module”). The “AFH - 7 day recall module” was tested against the same module but filled out each day by each household member under strict supervision over a period of 7 days. This was later considered as the “benchmark” against which the “AFH - 7 day recall module” was tested. The experiment found that the number of meals reported as well as the amount spent, collected through the “AFH - 7 day recall module”, were very close to those reported in the “benchmark”. This was considered an improvement compared to the way food consumed away from home was previously captured in HIES of the Pacific Islands countries, where only the amount spent in restaurants and bars was collected for the entire household (“one line expenditure” approach). The FAFH module was therefore introduced for the first time in the Pacific in the HIESs conducted in Vanuatu, the Marshall Islands, Kiribati, and Wallis and Futuna in 2019/20. It was found that the contribution of FAFH to dietary energy consumed in countries where the new module was adopted averaged 12% (except in Vanuatu), while the contribution of FAFH consumption was less than 3% of total DEC in countries using the “one line expenditure” approach⁶.

Despite the methodological improvement provided by the FAFH module described above, what is still unknown is by how much the amount of Dietary Energy Consumed Away From Home (DEC_{AFH}) estimated from this module differs from the “true” amount of dietary energy consumed away from home. The RMI experiment, in fact, did not have a real benchmark against which the module could be tested. As a matter of fact, neither the composition of meals nor the quantities consumed were surveyed; only information on the number and cost of the meals consumed was collected.

However, it is only by asking each respondent to report on all the products consumed away from home and their quantities that the amount of dietary energy consumed away from home can be accurately estimated. Since such information is difficult and costly to collect, alternative methods have been deployed to extrapolate DEC_{AFH} using an indirect method that combines information on the expenditure on meals consumed AFH collected in the HIES and the cost per calorie of the basket of food consumed within the household (hereafter referred to “in-house kcal cost”).

The “in-house kcal cost” is applied to the expenditure reported on AFH meals as:

$$DEC_{AFH}(kcal) = \frac{\text{Amount spent on meals consumed AFH (\$)}}{\text{in - house kcal cost (\$ per kcal)}}$$

This approach assumes that not only the expenditure on AFH meals are well captured in the HCES, but also that the meals purchased and consumed outside the household are composed of roughly the same ingredients as meals consumed within the household. Additionally, it assumes there is no margin for the food seller and no cost recovery for the rent and salaries required to run a food vending business.

⁵ Of note, most of the household survey conducted in the Pacific are Household Income and Expenditure Surveys (HIES) which is nationally representative household survey equivalent to HCES as they both collect information on food data.

⁶ Using data from the Pacific Food Consumption Database (Sharp et al 2024), we calculated the share of FAFH consumption in total dietary energy consumption for 15 surveys of the PICTs and found, using a simple mean, FAFH had a greater share in total energy consumption using the FAFH module than the one-line approach.

To address this second assumption, a “multiplier” is usually used to increase the FAFH cost per calorie by a standard percentage, to account for the higher cost of purchasing prepared food compared to cooking it. DEC_AFH is then estimated as:

$$DEC_{AFH}(kcal) = \frac{\text{Amount spent on meals consumed AFH (\$)}}{\text{in-house kcal cost (\$ per kcal)} * \text{multiplier}}$$

The literature does not provide a standard value for this multiplier, and the literature from the Pacific is limited, including only a standard value of 1.5 used in the analysis of the 2009/10 PNG HIES. Further analysis of the 2018 RMI experiment found a multiplier of 1.25 corresponding to the ratio between the expenditure on foods consumed outside the house and those consumed in the house⁷. Since then, and in the absence of a more direct estimate, a multiplier ranging from 1 to 1.25 on the cost of one in-house kilocalorie has been used when estimating DEC_AFH using expenditure on AFH meals collected in the most recent HIES conducted in the Pacific.

Furthermore, the “in-house kcal cost” approach is not standard worldwide. To estimate DEC_AFH using expenditure on AFH meals collected in HCES, FAO uses the median cost per calorie consumed in the house, estimated by area and income quintile. The World Bank uses a median cost per calorie consumed in the house by households belonging to the lowest percentiles (“flat cost”) adjusted by a multiplier to account for marketing margins and costs of running a food business, while SPC applies a multiplier to the median cost estimated following FAO approach.

To what extent the “in-house kcal cost” and the use of the multiplier are approximating the “true” value of DEC_AFH has not been assessed yet. A proposal to conduct an experiment to fill this methodological gap and to propose an alternative approach to the estimation of DEC_AFH was developed and submitted to the 8th Meeting of the Pacific Statistics Methods Board (PSMB) in October 2021. The PSMB agreed on the relevance of the FAFH project, noting that this has been an outstanding issue for PSMB recommendations on the methodology for poverty analysis (SPC, 2021).

The ‘Survey Experiment to Improve Data Collection and Estimation of Food Consumption Away From Home in Samoa’ (hereafter referred to as “AFH experiment”) was developed as an Innovative Experiment of the PACSTAT project. The experiment received financial support from the Samoan Government, the Pacific Community, the Australian Government, the Australian Centre for International Agricultural Research (FIS 2018/155), and the World Bank (P169122).

The experiment aimed to address two main questions:

- Test the in-house kcal cost approach to estimate DEC_AFH and assess the gap between the cost per calorie consumed in the house and the cost per calorie consumed away from home;
- Propose an alternative method to collect FAFH consumption in a more comprehensive way and obtain a direct measurement of DEC_AFH.

Through the experiment, a few more questions could be addressed, such as:

- Is the indirect approach used to estimate dietary energy consumed away from home a valid approach?
- Is the 7-day recall module administered at the individual level a good instrument to collect meals consumed away from home by meal event?
- What are the main aspects of food consumption away from home in Samoa?

The experiment was implemented by SPC in collaboration with the Samoa Bureau of Statistics (SBS) from January 2023 to December 2023, which also corresponded to the period the Samoa 2023 HIES was conducted. The design of the Samoa 2023 HIES was aligned with the WB/FAO guidelines on food data collection, which was one of the prerequisites for the experiment.

⁷ See unpublished note: “Outstanding questions for PSMB”, K. Himelein. WB. 2019.

2. Methodology

Description of the experimental design

The survey experiment described herein involved two representative sub-samples of households (A and B), receiving 3 modules each: one module (Module 1) to gather information on the socio-economic, geographic, and demographic characteristics of each household member; one module (Module 2) to collect household in-house food consumption data; and one out of two modules (Module 3 or 4) to collect data on FAFH consumption, with Module 4 serving as the benchmark.

The two sub-samples were representative of the population in terms of age, sex, income, and region, so that the information extracted from one sub-sample could be expanded to the entire population using household sampling weights⁸.

The figure below provides a detailed description of the research design.⁹

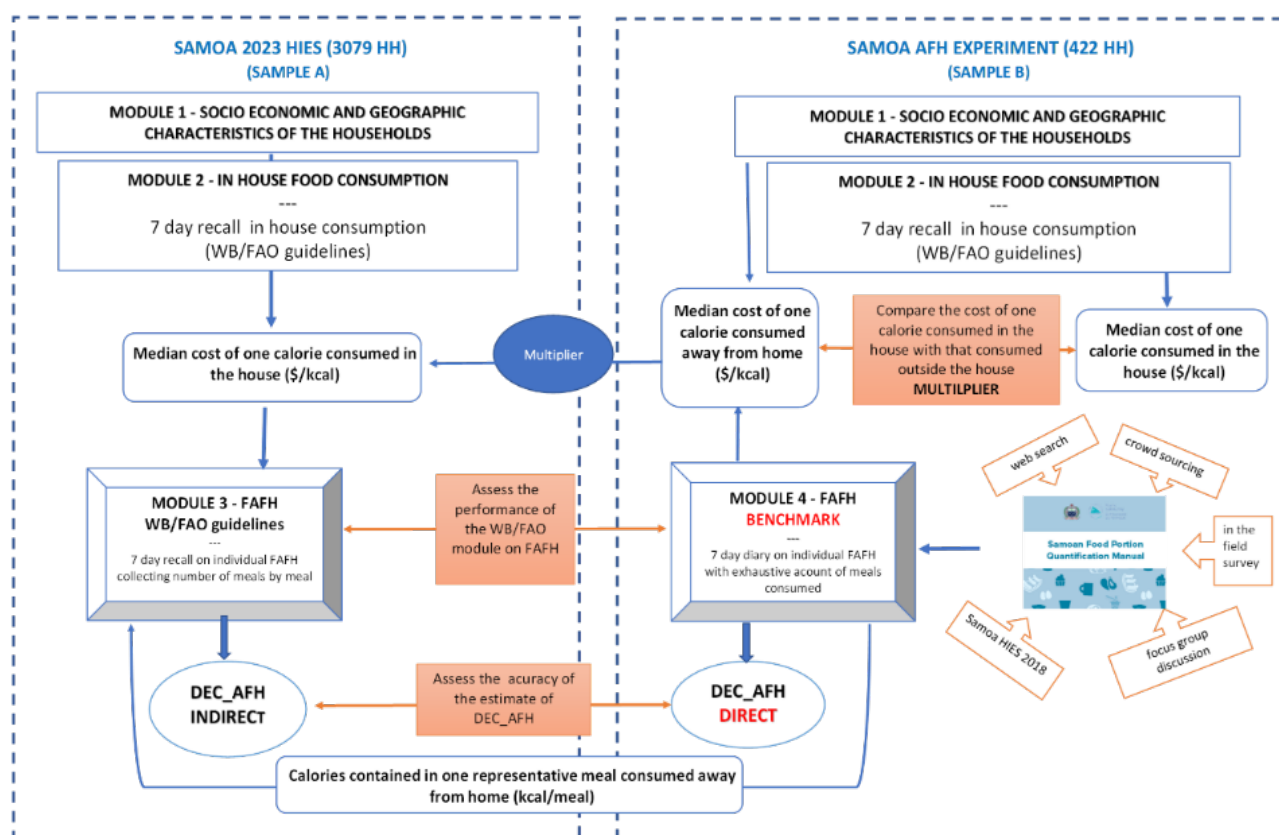


Figure 1. Description of the experiment

The four survey modules and their use in the experiment are described below:

- Module 1 – Socio-economic characteristics of the household members**

This module collected demographic and socio-economic characteristics of the individuals, such as age, sex, education level, income category, economic sector, geographic location, etc. The information collected in this module was used to estimate the differences in the cost per calorie by population groups.

⁸ In both samples, sampling weights were constructed based on characteristics of the household head age, sex, education, economic activity and geographical location using 2021 census.

⁹ Number of households in the brackets was tentative at the time the project document was designed.

- **Module 2: In-house food consumption – 7-day recall**

This module aimed to estimate the average amount of dietary energy consumed in the house and to derive the in-house cost per calorie; it corresponds to the 7-day recall module recommended in the WB/FAO guidelines. In this module, the person responsible for preparing meals for in-house consumption is asked to report on the quantities of each food consumed by the entire household over the last 7 days and the corresponding cost.

- **Module 3: Module on Food Away From Home – 7-day recall**

This is the module recommended by the WB/FAO guidelines and adopted in the latest HIES conducted in the Pacific, which uses the 7-day recall method to collect food away from home consumption expenditure. In this module, each household member is asked to report on the number of meals (breakfasts, lunches, dinners, snacks, hot beverages, and drinks) consumed away from home over the last 7 days, by meal event, and the respective amount of money spent (or the estimated value of food received in kind, in case of no monetary expenditure)¹⁰.

- **Module 4 – Module on Food Away From Home – 7 day diary - Benchmark**

This module collects accurate information on each meal consumed away from home per day over the last 7 days. Each household member is asked to fill out a diary to report on the exact foods consumed away from home with the support of a manual on food portion quantification, which was provided to the household. In case a food cannot be found in the manual, the respondent is requested to describe the food consumed, its main ingredients, and the size of the portion using standard food measurement units provided in the food quantification manual. Additionally, information is also collected on the place of consumption (e.g., restaurant, bar, school, work, etc.), the actual or estimated amount of money spent, the day of the week when it occurred – to capture weekly patterns of FAFH consumption, and the type of meal. (see screenshot of the diary in Annex 1).

Data collection methods (sample selection, diaries, observations)

The Samoa 2023 HIES and FAFH Experiment were implemented concurrently over 36 weeks from March to December 2023. A total sample of 3,867 households were selected to participate in either the FAFH experiment (486 households) or the HIES (3,381 households).

The HIES sample was selected using a two-stage stratified sampling approach, which aimed to minimise relative sampling error with respect to estimates of total household consumption, by strata of urban (i.e., Apia, with 27% sample allocation), North-West Upolu (33%), Rest of Upolu (20%), and Savai'i (20%). The sample was designed to consider non-response, based on the non-response rates (by region) measured in the Samoa 2018 HIES. The households selected for the experimental arm were allocated proportionally and respectively to that of the HIES sample, by strata.

As all households selected for the FAFH experiment also had to fill the modules from the core HIES, the fieldwork required additional organisation to administer the Samoa 2023 HIES and the experiment simultaneously. Households selected for the FAFH experiment received a diary (see section on tools and instrument) which required numerous visits for monitoring purposes, while the HIES households only required one visit to complete the entire interview, including the data required to estimate both in-house and away from home food consumption.

Field organization

The FAFH experiment followed the same schedule and field organization as the Samoa 2023 HIES.

¹⁰ In this report, whenever we will refer to expenditure on meals AFH, we refer to both the direct purchase and the estimated value in case the meal was received as gift.

Rounds

- The Samoa 2023 HIES was divided into 36 rounds, with some breaks to accommodate disruptions to the schedule, such as holidays. The Samoa 2023 HIES was originally designed to be administered over a quasi-12-month period to account for seasonal fluctuations in income, expenditure, and consumption patterns, from January 2023 to November 2023. However, due to extenuating factors, the enumeration was delayed by a month and therefore started in March 2023 instead.
- A round of collection corresponded to a 1-week period during which each team (2 interviewers and 1 supervisor per team) had to survey 14 households (i.e. 2 households per day per team, or one per interviewer). These 14 households were located within the same “Enumeration Area” (EA). In total, 84 households were interviewed per round by the six HIES teams. Accordingly, each interviewer interviewed 7 households per round.
- For the FAFH experiment, each round had a duration of 2 weeks, hence the total number of rounds was 22. In each 2-round period, a total of 24 households were interviewed (6 interviews per team, 4 teams).

Field operation

A total 6 teams were in the field, operating as follows:

- Teams 1, 2, 3 and 4: Each team comprised of three interviewers and one supervisor. Two interviewers were responsible for conducting interviews for the HIES, with each enumerator completing seven households per round). The third enumerator focused exclusively on households selected for the FAFH experiment, completing six household interviews over a two-week period.
- Teams 5 and 6: These teams, each consisting of two interviewers and one supervisor, were exclusively responsible for conducting HIES interviews.

Interviews were scheduled within geographical strata for each team to capture seasonal variations in household consumption. In fact, both food and non-food consumption patterns, as well as income generating opportunities, fluctuate throughout the year. If all the households in a specific region were surveyed during one season while another region was surveyed in a different season, results would not be fully comparable. To ensure both spatial and temporal comparability, all the 4 regions/strata (Apia Urban Area, North West of Upolu, Rest of Upolu, and Savai'i) were covered in each round.

Team schedule by enumeration area (EA)

The FAFH experiment required an additional week in each EA due to the need for a 2-week monitoring period for household food diaries. As a result, the FAFH enumerator in Teams 1 to 4 remained in the same EA for 2 weeks. The other two team members stayed in an EA for only 1 week before moving to another HIES EA the following week. In the third week, all three team members regrouped again in an EA where both HIES and FAFH households were surveyed.

Mode of data collection

Interviews were conducted using a Computer Assisted Personal Interview (CAPI) questionnaire administered on a tablet, in which the enumerator could select the 7-day recall for all households selected for the HIES, while this option was removed for all households selected for the FAFH experiment. The paper diary used for the FAFH experiment was transcribed into the CAPI application by the enumerator during the last visit to the household.

The FAFH diary (paper questionnaire) was printed in both English and Samoan. Similarly, the manual on food portions quantification was printed in a single version, including both languages.

To have comparable recall periods for both in-house and away from home food consumption for the experiment arm, the HIES (CAPI) questionnaire was completed by household members on Day 7 of the interview, according to the schedule below:

- Visit 1: The enumerator delivered the food portion quantification manual and a number of FAFH diaries equal to the household members currently living in the household, for those selected for the FAFH experiment. The number of diaries to be left was confirmed verbally with the respondent. The household head and other members of the households were instructed on how to fill in the diary and use the food portion quantification manual, using the example included in the paper diary. Arrangements were then made about the timing of Visits 2, 3 and 4. Respondents were encouraged to complete the individual diaries autonomously, without waiting for the next visit by the enumerator.
- Visits 2 and 3: The FAFH enumerator visited the household to review the FAFH diary, helping the household members overcome the major challenges in filling it out and referencing the food portion quantification manual to prevent under-reporting or mis-reporting of food consumed AFH.
- Visit 4: The enumerator helped the household fill in Days 5 to 7 of the diary. The enumerator also completed the HIES questionnaire with the household head using the tablet, including transcribing the FAFH information reported in the paper diary into the CAPI system. As this visit carried the heaviest workload for both the enumerator and the household, the interview day and time were planned in advance with the household head to avoid conflicts with busy or unavailability.

NOTE: no interviews took place on Sundays. Therefore, visits falling on Sunday (and on following days) were moved back by one day. This meant that the fourth visit occurred on Day 8 instead of Day 7. However, the food reported in the diary was still meant to cover the period from Day 1 to Day 7. It is believed that some of the food consumed during To'onai¹¹ that occurred on Sunday might not be fully captured in the experiment.

Treatment of refusal

When households refused to fill in the diary, the enumerator proposed the following alternatives:

- Assisting the household in completing the diary on Day 1, 3 and 5 without leaving the diaries with them.
- The FAFH household was replaced as HIES household and complete only the HIES questionnaire on the tablet (without the FAFH diary). The next selected HIES household in that EA would then be assigned as a FAFH household and required to complete the diary.
- Only in cases where the household still refused to participate in either the HIES or the FAFH experiment was it considered as “lost” and not interviewed. However, this option was avoided as much as possible.

In total, only 5 households persistently refused to participate in the experiment.

Testing of the material and training of enumerators

Due to time constraints, neither cognitive testing nor a full pilot of the diary could be conducted. Instead, enumerators received an abridged 3-day training in November 2022, followed by pre-survey testing of the instrument, including experiment tests, in December 2022 and January 2023. Refresher training sessions were held in late-January, just before data collection began. Enumerators were provided with digital copies of the training materials, field manuals, questionnaire manuals, HIES schedules, and maps for reference while working in the field. Follow up training and data reviews were conducted quarterly for quality control purposes, and to address any operational challenges.

Final sample for analysis

Out of the 3,867 households selected for both the HIES and FAFH experiment, interviews could not be conducted for 366 households (9.5%), mainly because the house was vacant. These households were not replaced, resulting in a final valid sample was 3,501 households, of which 422 were selected for the FAFH experiment.

¹¹ The To'onai refers to a feasting event occurring on Sunday after Church service. During To'onai, members of the Church or family are bringing foods to be shared and consumed by all people present.

Table 1. Distribution of the total households between the HIES and the experiment

Interview type	Freq.	Percent	Cum.
HIES	3,079	87.95%	87.95%
Food Away From Home Experiment	422	12.05%	100.00%
Total	3,501	100.00%	

Table 2. Main reasons for non-response

Reason for non-respondent	Interview type		Total
	HIES	FAFH Experiment	
Persistent refusal	5	0	5
Household away during	6	1	7
Household not available	6	2	8
Vacant house	195	46	241
Cannot contact the household	1	0	1
Other reason	89	15	104
Total	302	64	366

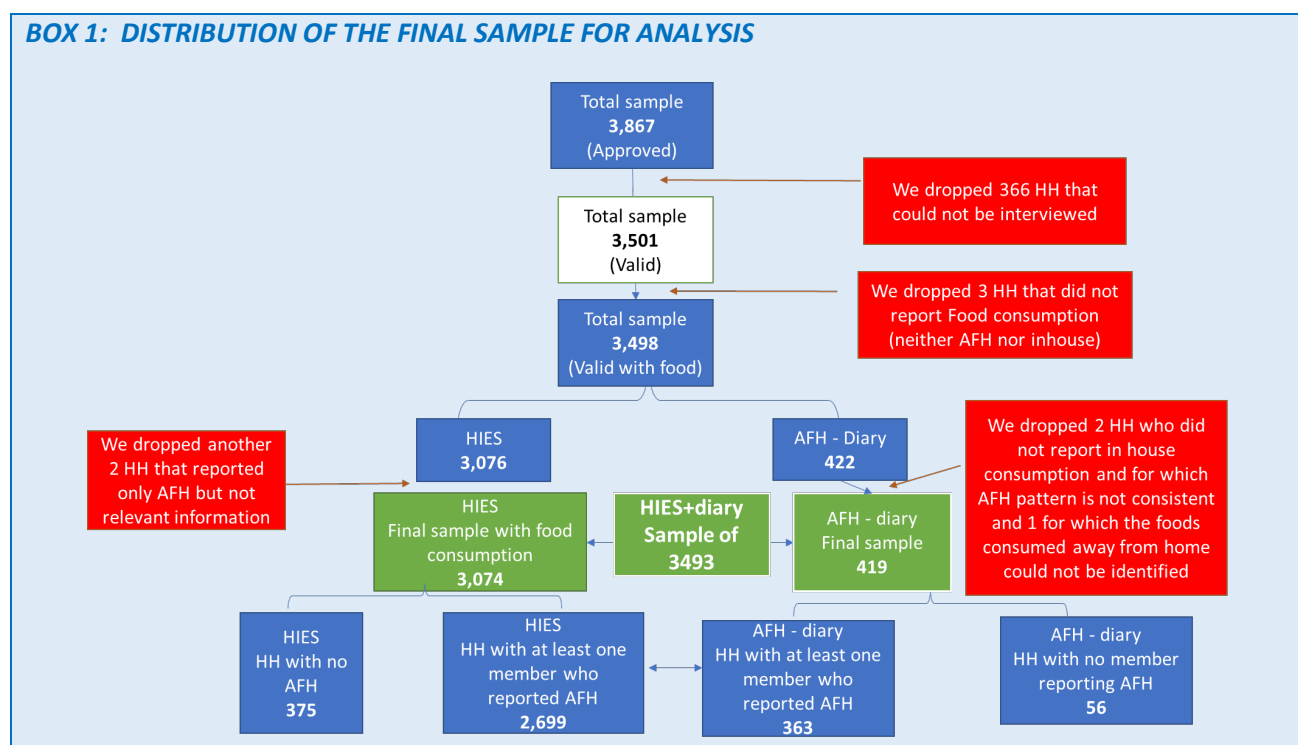
After dropping two households from the FAFH sample that did not report in-house food consumption and had inconsistent away from home consumption, the analysis of the FAFH experiment data is based on 419 households, of which 363 reported away from home¹² consumption. A total of 3,074 households were considered for the HIES analysis.¹³



¹² This number refers to the number of households for which at least one member filled the FAFH diary.

¹³ Of note, out of the 3,074 for which an average estimate of household per capita per day dietary energy consumption was obtained, 3 households were excluded from the final analysis on food consumption patterns in Samoa due to extremely low or high food consumption.

BOX 1: DISTRIBUTION OF THE FINAL SAMPLE FOR ANALYSIS



Observations

- **Sampling and Mapping:** An original sample of households was selected for the experiment. The sample was provided by SPC staff based on a map using the latest GPS coordinates available. The sample was designed to be representative of the population of Samoa by urban and rural area and the 4 strata. However, problems arose due to the late delivery of maps from the Census office of SBS, which delayed all sampling and mapping activities. This delay prevented thorough field testing of the questionnaire and did not allow sufficient time to develop maps.
- **Data collection:** During the training of enumerators, a light testing of the diary was performed within SBS. It was found that the diary was too lengthy and difficult to complete despite the detailed instructions, and this was further confirmed by enumerators during the initial days of data collection. Indeed, many respondents either refused to fill out the diary or lacked the time to complete it daily. Although the data collection process was intended to involve each member of the household filling out a daily diary – and checked every other day by enumerators, in practice, it often turned into a two-day recall process, with enumerators assisting respondents. For households that had difficulty completing the diary, SBS instructed them to make a list of FAFH consumption by meal event each day to facilitate recall during the enumerator's visit when the diary was completed.
- The food consumed from potluck gatherings (To'onai) proved difficult to be captured. Furthermore, data collection did not occur on Sundays, so this consumption may not have been well captured in the diary.
- When respondents could not find the meal in the food atlas, they were asked to provide details on the composition of the meal or cooking methods. However, in most cases, only the name of the food was provided without further details, slightly affecting the estimation of nutrient content of the food consumed. More issues related to data entry are further discussed in the next section on data analysis.
- Completion of the diary was complicated, making it more a recall process when the field staff assisted the households with diary completion. However, based on this experience, households seemed to prefer the recall method over the diaries.
- The two main tools used for the experiment were the diary and the food atlas. The development of these two instruments required extensive planning, consultations, and the involvement of experts working on similar issues. An expert group consisting of SPC and SBS staff, along with experts in survey

design, food data analysis, and nutritionists, was created. Various technical meetings were organized, and the diary was primarily developed through consultations with the expert group. These technical meetings proved to be very useful and helpful in the development process, not only to adjust and refine the tools but also to exchange knowledge and experiences on innovative experiments in this field.

Tools and instruments used

Two main instruments were developed for the FAFH experiment: the manual on food portion quantification and the diary. The development of these two instruments started in February 2022 and ended in December 2022 through the implementation of various activities and expert consultations.

Development of the manual on food portion quantification

Four main activities were implemented to develop the manual:

- Establishment of the FAFH meal sampling frame
- Cooking and photography sessions
- Estimation of the nutrient values of the foods
- Compilation of the manual

Establishment of the FAFH meal sampling frame:

The food quantification manual required a list of meals commonly consumed away from home in Samoa, data on the nutrient composition of those meals for analysis, and photographs of standard portions of those meals for data collection purposes. Through a general search and discussions with project stakeholders, business registers were obtained; however, no lists of Samoan meals were found. Therefore, we undertook five activities to establish the meal sampling frame, including: i. Web search, ii. Data extraction from Samoa's 2018 HIES, iii. Discussion groups, iv. Physical enumeration, and v. Crowd sourcing.

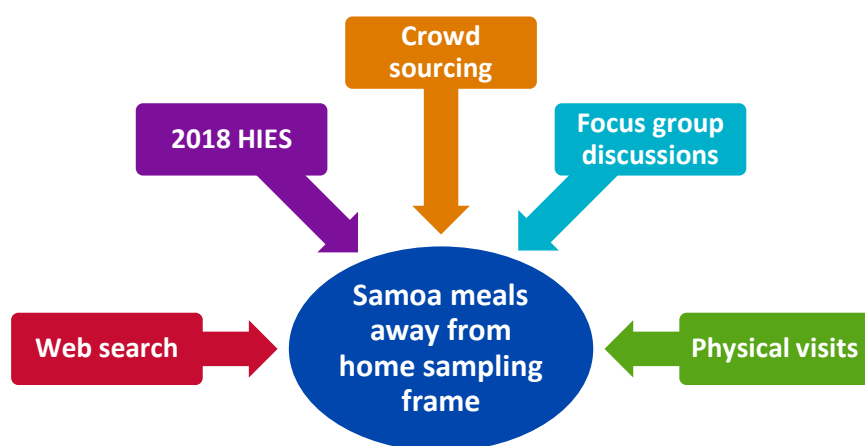


Figure 2. Main activities involved in the development of the Meals Away from Home Sampling Frame

1. Web search

This activity was relatively low cost and involved browsing the web to gather available information on food establishments in Samoa (menus, locations, costs) as well as blogs or videos featuring recipes for traditional Samoan foods.

From this activity we compiled a list of:

- 123 hotels with their location, GPS coordinate when available, and website;
- 199 food establishments in Upolu and Savai'i with information on their location (GPS coordinates when available), main menus, types of establishments (pizzeria, bar, restaurant, high-end restaurant), webpage and some information or pictures of meals served;

- 63 recipes of traditional Samoan foods.

The web search initiative was efficient in identifying food vendors; however, the digitisation of menu items was a labour-intensive process. It also primarily collected information on meals available from formal establishments, which featured menu items seemingly targeting tourists, and therefore missed informal food vendors, caterers, school meals, and the anecdotally significant source of FAFH consumption—the Samoan To'onai (Sunday lunch at the church or with relatives).

2. 2018 HIES

Description of all food and beverage items (from COICOP Divisions 1 and 2) were extracted from the dataset of Samoa's 2018 HIES. This was very easy and low cost activity that only required access to the food data collected in the 2018 HIES.

3. Focus group discussion

To gather information from a variety of food vendors, consumers, and local experts, a meeting of Samoan representatives of food vending establishments, schools, churches, elderly, youth, and representatives from Government Ministries such as of Health and Education, hosted by SBS on 22 June 2022, aimed to gather input on the experiment framework and aid in compiling the meal sampling frame. The discussion addressed key questions, including:

- Where do Samoans typically consume food away from home (e.g., at food establishments, school, house of friends and relatives)?
- What are the FAFH consumption habits in urban and in rural areas (e.g., frequency of consumption, weekly/annual patterns, places of consumption)?
- What are the main meals consumed away from home in urban and in rural areas (e.g., list of meals, traditional foods)?
- How to approach food establishments and street vendors to obtain recipes, and when is the best time to visit them?

A total of 37 people participated in the meeting. Small discussion groups of around 6 people were formed to discuss these questions, followed by a plenary discussion where each group summarised their discussion. The list of food vendors and meals from the web scraping and 2018 HIES was reviewed and updated, with obsolete items removed. Minutes of the discussions were recorded and reviewed by the project team after the conference.

4. Crowd sourcing

It was identified that some Samoan foods, commonly consumed during significant religious, cultural, or celebratory events (especially on Sundays), had not been covered by the initiatives undertaken so far. As a result, a web-based data collection instrument was developed using Survey Solutions software, accessible via a quick-response (QR) code. A data collection campaign, 'Snap Your Meal' (or 'Snap your Meaai'), was launched in both in English and Samoan to capture descriptions and photos of meals consumed away from home by the general public.

The data collection instrument was designed to capture two general types of meals: single-serve and buffet style, with the latter being common practice during cultural events. The single-serve approach was relatively simple where the respondents were asked to identify the location where they consumed the meal and upload a photograph of the meal they consumed, including beverages and side dishes. The buffet-style approach asked for photos of each plate of food consumed and a photo of the entire buffet with descriptions of each dish. Additional questions were asked to all respondents for both the single-serve and buffet-style meals, requesting them to list the main meals, beverages, and locations where they eat food away from home.



Figure 3. Flyer for snap your meal initiative

The data collection campaign was advertised on social media, including the Facebook page of SBS, in the email footer of SBS staff, and through printed flyers displayed in busy areas of Apia town, Savaii and shops around Upolu. Small cash prizes were offered as incentive for participating in the data collection initiative, which ran for 2-weeks, from 29 September to 11 October 2022.

Following the campaign, data were downloaded and analysed using Stata SE/15.1. The crowdsourcing resulted in 142 responses from individuals from 9 to 61 years, with a mean of 32 years. More than one-third of respondents were based in an urban area, another one-third in peri-urban areas, while 15% were located in rural Upolu and 6% in Savai'i.

Almost half of the meals were consumed at a workplace events, nearly one-third at commercial establishments, 10% in a public area (e.g., parks, seawalls, or cars), and the remainder at the homes of friends or relatives, on public transport, schools, and Churches.

Single servings were the most common meal type, accounting for 86% of all reported meals, with the remaining 14% were buffet-style meals. Although not interesting to comprehensively describe, the results of the open-ended question requesting the name of the meal consumed, items on the buffet, and beverages provided important insight into FAFH consumption patterns. For example, around one-quarter of meals often consumed away from home were chicken-based, including chicken and chips and chicken curry. Along with fish and chips, pork buns, noodles, fried rice, sandwiches, and burgers, these meals comprised nearly two-thirds of meals reported as being frequently consumed away from home.

5. Physical enumeration

It was identified that the rural island of Savai'i, home to around one-quarter of the Samoan population, as well as food available from more traditional or local outlets—such as vendors at food market and roadside stalls, were not well covered by the web search or crowdsourcing activities. To address this gap, the project conducted a physical enumeration of Savai'i, during which the SBS's Economic Statistics team, responsible for the HIES and prices collections, visited all restaurants, cafes, and food vendors on the island, including those at the food market, to photograph menus and available food items. Additionally, the main food markets in Samoa's capital, Apia, were visited to collect the same information.

Final FAFH meal frame:

Through activities (1) to (5), a FAFH meal frame was successfully established. The foods listed in the frame were then categorized based on their method of nutrient data collection, as follows:

- Composite meals that are to be prepared with detailed collection of ingredients, and photographed in numerous portion sizes (e.g., fish and coconut milk, chicken noodles, turkey tail curry).
- Composite meals that are to be purchased and photographed in their purchased form (e.g., a burger and chips, chicken skewer), sometimes with portioned items such as a slice of pizza or piece of a chicken. These meals were deconstructed and individually weighed but not prepared.
- Snacks purchased and presented in their whole form (e.g., a packet of chips, noodles, a doughnut, or an ice-pop).

This classification was determined by the authorship team on a case-by-case basis, considering available nutritional information.

A total of 113 FAFH meals or meal components were finally selected for the collection of nutrient data and inclusion in the food quantification manual. 80% of these meals were identified as composite meals to be cooked with the collection of nutrient information, 19% were foods to purchase and deconstruct, and 1% were meals to both purchase and prepare.

Cooking and Photography Session:

A strict protocol was developed for the photography session based on international guidelines^{14,15}.

Drinking glasses, mugs, bowls, and spoons to be used as reference measurement units were selected by SBS based on their knowledge of which items are commonly available and used by households in Samoa. These reference items were included in the food quantification manual with corresponding measurement units in centimetres to assist in interpreting portion sizes during data collection.

A commercial kitchen and chef were contracted to prepare the meals and provide a space for photography. A professional photographer was hired, and standard plates, bowls, and digital scales were procured to assist with preparation of the meals and weighing of ingredients. Standard protocols were established for photography session, including consistent distance, angle, lighting, and background.

The selection of portion sizes to display for each food item needed to consider the number of portions, the interval between portions, the range of amount available, and other aspects such as typical portion sizes for that food among the Samoan population. A sufficiently large number of portion options was displayed for each food item, depending on the typical range of consumption for that food. For example, a hamburger, is typically eaten as an individual item, and the quantity consumed depends on the size of the hamburger itself, so photographs were taken of burger meals in different sizes. Conversely, the same pizza or portion of chicken curry is more dependent on the consumer, so multiple options of pizza slices or curry portions were displayed, covering a range covering from small to large portions. It was important that the portion increments were not too large to lead to errors in estimating the amounts consumed, but at the same time, the difference in the portion or unit sizes depicted needed to be visually discernible to the respondent (Vossenaar et al. 2020). The recommended number of photos was six (Crispim et al. 2014), however, this was not strictly adhered to.

The general process was as follows:

¹⁴ See Crispim S, Nicolas G, Knaze V, Freisling H, Slimani N. Reference guidelines for the preparation of phot Steps for Recipe Disaggregation in Food Consumption Surveys for food portion quantification in EPIC-Soft. Lyon: International Agency for Research on Cancer. 2014.

¹⁵ See Vossenaar M, Crispim SP, Lubowa A, Deitchler M, Moursi M, Arimond M. Guidance for the Development of Food Photographs for Portion Size Estimation in Quantitative 24-Hour Dietary Recall Surveys in Low-and Middle-Income Countries. Washington, DC and Curitiba, Brazil: Intake–Center for Dietary Assessment; 2020.

- Ask the chef to identify the minimum and maximum portion sizes, often ranging from a soup spoon to a full plate.
- Identify intermediate portion numbers and sizes to be displayed, with the serving size increment often calculated as the difference between the maximum and minimum portion size, divided by the number of portions minus one.
- Visually confirm the usability of the selected portion sizes for each food item and adjust the portion sizes, if needed.

Each portion was plated and weighed using the digital scales, with the plate weight subtracted. Data were recorded on paper collection forms and later entered into an Excel database. Where there were inedible components, they were removed, and records of the weights of inedible components were similarly kept. Protocols were prepared for the time between food preparation and photography, including methods to preserve the colour and integrity of the food, and for the placement of food portions. This included not heaping and filling the plate progressively from left to right for smaller to larger portion sizes.

The same photographic angle was used for all portion sizes of a given food item, and the distance between the camera and food was consistent for all photographs. Photographs were taken with good lighting and at high resolution to allow for large, high-quality prints, with an unobtrusive, neutral, light grey background. For food items where visualizing depth was necessary to quantify consumption, the food item was turned at an angle (e.g., for a slice of cake). The same fork and knife were placed as a reference alongside the plate.

The food portion quantification manual

The final document consists of 82 foods with 420 portions, as well as 27 household measurements. The food quantification manual was designed to include photographs of measurement units, with foods grouped by their general categories (e.g., fish and seafood, meat-based, pasta and noodle, fruit and vegetables, soups, dessert and sweets, and composite meals).

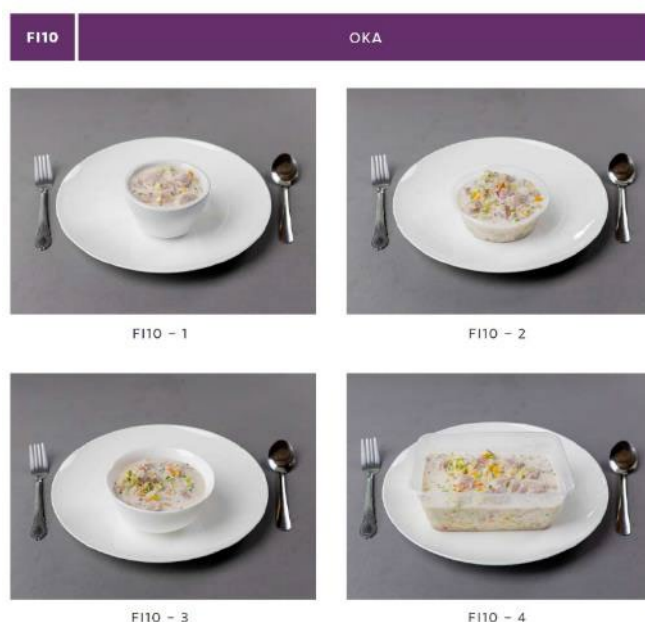


Figure 4. Example of portion picture in the manual

The nutrient composition for each portion was determined by linking it to available regional food composition table or database. In some cases, recipe data disaggregation was used.

For recipes, the composition and contribution to nutrient intake of each ingredient must be estimated with the highest possible degree of accuracy. Ideally, all recipes should be disaggregated from scratch, by preparing and weighing all ingredients before and after preparation. However, due to time and resource constraints, it was not

feasible to prepare and weigh all the foods selected for the manual. Instead, post-recipe disaggregation consisting in estimating measurement rather than actual weights was adopted in some cases, using the Direct Method proposed by Greenfield & Southgate¹⁶. Briefly, a standard recipe was identified, followed by the identification of ingredients with their method of preparation and their proportions, in relation to the total weight. After, the nutrient composition of the food was considered based on the proportions and values available in the food composition tables.

The core database used to link the foods with the composition data was the Pacific Nutrient Database, followed by the Australian database (AUSNUT 2011-13) and the New Zealand database (NZ Food Files 2014). In a few cases, data from the USDA Food Composition Database (USDA) and the Brazilian Food composition database (TBCA) were also used.

Table 3. Example of a table with the nutrient composition of each food portion in the manual

Code in the Atlas	Food name in Samoan	Food name in English	Weight (g)	Energy (Kj)	Energy (kcal)	Protein (g)	Fats (g)	Fiber (g)	Carb. (g)
FI10 – 1	Oka 1	Oka 1	232	1,096	263	16.8	17.5	1.5	8.9
FI10 – 2	Oka 2	Oka 2	267	1,262	303	19.3	20.2	1.8	10.2
FI10 – 3	Oka 3	Oka 3	301	1,422	342	21.8	22.7	2.0	11.5
FI10 – 4	Oka 4	Oka 4	624	2,949	708	45.1	47.1	4.1	23.8

A large food composition database was then created corresponding to the nutrient composition of 82 meals from the manual, along with 130 foods that were reported in the diary but not pictured in manual (example Coca Cola, ice pop). For the latter, the nutrient composition of the food was obtained from reference food composition databases when the food was well described. When the food referred to a prepared meal (such as fried mackerel), nutrient composition was estimated using recipes sourced from the web search activities or the SBS.

In the manual on food portion quantification, the energy (expressed in kcal and kj), as well as the quantity in grams of proteins, fats, carbohydrates, and fibers, is provided for each food portion, as along with its weight in grams. The manual on food portion quantification is publicly available on the SPC and SBS websites.¹⁷

3. Results

Data summary and preparation of the working dataset

The diary was administered to 422 households, but 3 households were excluded from the analysis: 2 households were removed because they did not complete the in-house module, and their food consumption pattern were inconsistent with a household that exclusively eats away from home; and 1 household because the two meals consumed away from home could not be identified.

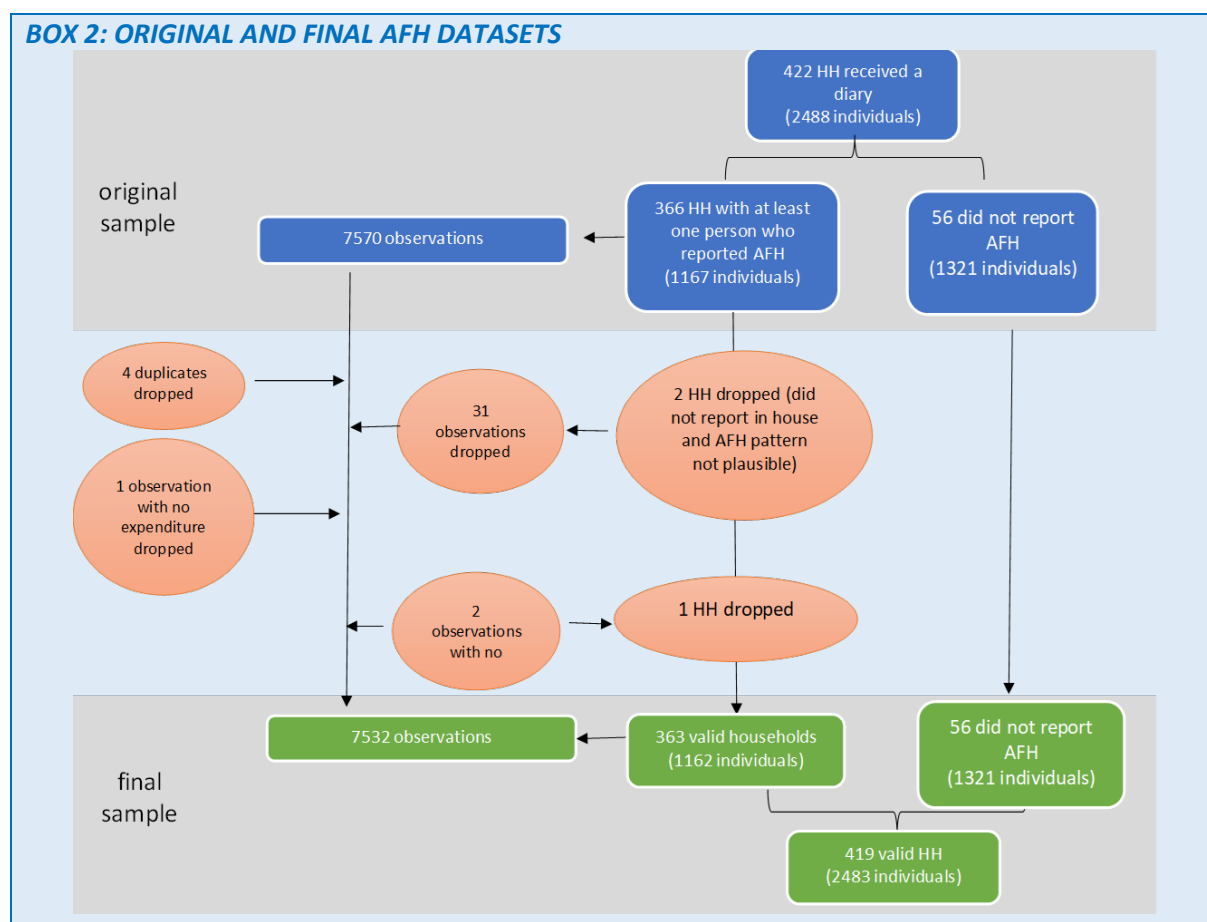
A total of 7,570 food consumption events were reported in the diary, but 38 observations were dropped¹⁸, bringing the final sample to 7,532 food consumption events corresponding to 1,162 individuals from 363 households.¹⁹

¹⁶ See Greenfield, H. and Southgate, D.A.T. Food composition data: production, management and use. London: Chapman & Hal, 1992. 243p.

¹⁷ https://sdd.spc.int/digital_library/samoan-food-portion-quantification-manual

¹⁸ 4 observations corresponding to duplicates, 2 observations corresponding to foods that could not be identified, 1 observation corresponding to a quantity for which we did not have the expenditures and 31 observations corresponding to 2 households that did not report in house food consumption and their away from home consumption alone was not consistent with what we would expect from individuals eating exclusively away from their home each day for 7 days.

¹⁹ Which means that none of the people belonging to 56 households consumed away from home the last 7 days.



Lunch was the most frequently reported meal, accounting for 63% of consumption events, followed by breakfast and snacks.

Table 4. Distribution of meals by meal events

Meal type	Freq.	Percent	Cum.
Breakfast	1,646	21.85%	21.85%
Lunch	4,736	62.88%	84.73%
Snack	659	8.75%	93.48%
Dinner	491	6.52%	100.00%
Total	7,532	100.00%	

A total of 414 different FAFH meals were consumed, of which 313 were portions identified in the food quantification manual, corresponding to 5,612 observations.

Approximately 75% of the food consumption events reported in the diary were associated with a food portion. For the remaining food consumption events where a portion picture was not available (that is 1,920 food consumption events corresponding to 101 different food products), the quantity was reported either in a unit of measurement for which a picture was available in the food portion quantification manual (885 food consumption events²⁰), or in a unit of measurement further described by the respondent (for example 330ml can of soda).

²⁰ Original number of quantities was 893, but 3 quantities corresponding to a food we could not identify were further dropped.

interview_key	hhroster_id	ay_roster	meal_type	fahf_list	portion_picture	measure_picture	unit_measure	other_unit	qty_consum	meal_locat	meal_purc	meal_cost
86-48-85-59	6	1	Lunch	soft drink	999	M4-1						
94-71-96-32	7	4	Lunch	ipu kofe	999	M2-5						
17-65-57-35	2	5	Lunch	can soda	999	999	Standard	330ml				
18-81-06-19	4	1	Snack	ice pop	999	999	Standard	each				
53-97-41-86	2	2	Breakf	pakua moa kao	999	999	Standard	each				
47-79-91-00	1	2	Breakf	coffee	999	M2-5 (3/4)						
47-81-79-91	4	1	Lunch	pancake	B13-4							
83-97-79-68	1	3	Breakf	Round Pancake	B13-4							
62-90-36-19	11	5	Breakf	panikeke	B13-4							
31-75-55-13	4	4	Breakf	Pancakes	B13-4							
69-20-51-08	2	2	Lunch	pancake	B13-4							
15-79-40-08	2	2	Lunch	Pagikeke	B13-4							
31-75-55-13	5	3	Breakf	Pancakes	B13-4							
75-92-25-45	18	3	Breakf	pancake	B13-4							
68-03-56-27	4	4	Lunch	Panikeke	B13-4							
99-15-01-22	1	4	Breakf	Round Pancake	B13-4							
15-79-40-08	2	4	Lunch	Pagikeke	B13-4							
77-91-48-54	3	3	Lunch	panikeke	B13-4							
42-38-84-72	5	3	Breakf	pancake	B13-4							
20-51-84-87	2	3	Breakf	round pancakes	B13-4							
17-49-72-21	2	3	Breakf	pancake	B13-4							
93-86-92-30	4	5	Breakf	pancakes	B13-4							
15-79-40-08	2	1	Lunch	Pagikeke	B13-4							

Figure 5. Structure of the dataset

The main steps to convert all the quantities reported in the diary into estimates of dietary energy (i.e., kilocalories) were:

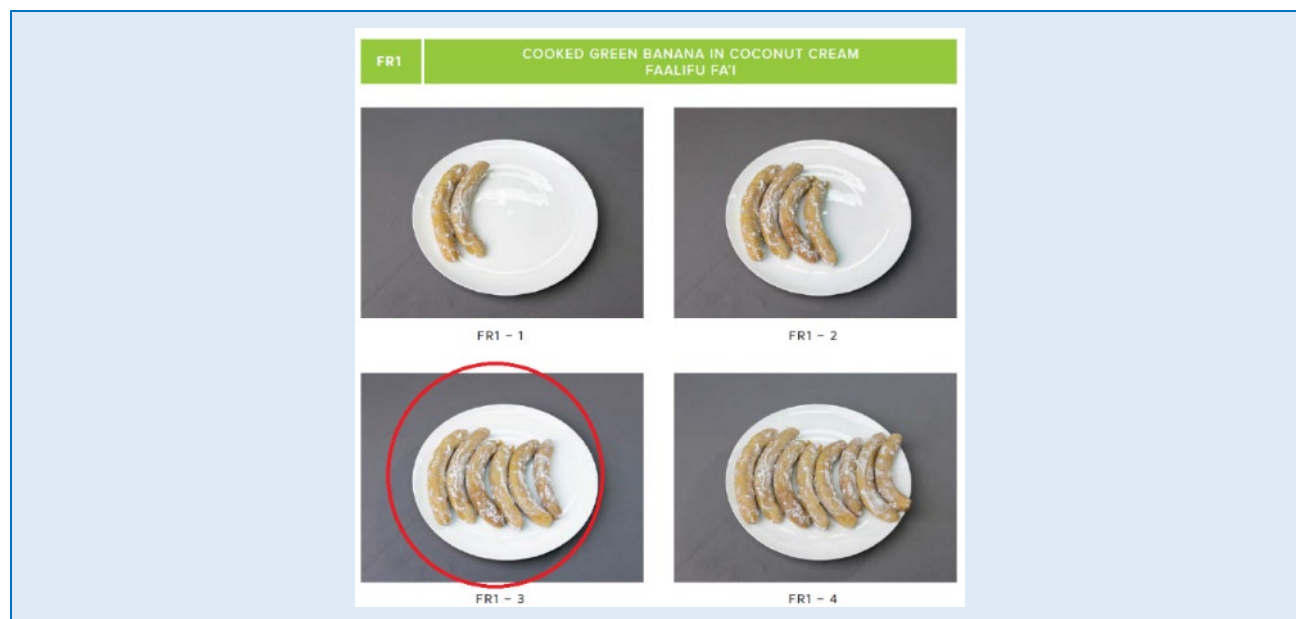
1. Edit the data

An important phase of data cleaning took place throughout all the stages, from data collection to data preparation. Since a proper pilot of the diary could not take place, a preliminary assessment of the data was performed in Mid-April 2023, a few weeks after the start of data collection. Based on this assessment, further instructions were provided to enumerators to improve the harmonization of codes and the description of foods when they could not be found in the manual.

Despite instructions, additional work was needed at the end of data collection to further harmonize all codes and the name of the foods.

BOX 3: MAIN ISSUES OBSERVED IN THE DATA COLLECTED

- The same product was reported, but different portion codes were used (e.g., “biscuits” entered as “B12-2A” or “B12-A,” with the latter being the correct code).
- Non-existent codes were entered (for instance “C01-3” instead of “CO1-3”).
- Food descriptions were in Samoan and needed to be translated into English to ensure that the code and all corresponding information were correctly allocated.
- The same code was assigned to two different products (instance e.g., “CO1-3” was used for both “BBQ” and “cup of water”).
- The codes corresponding to measurement pictures were entered in the field corresponding to portion pictures.
- The same product was recorded under different names (e.g., “bottle of water”, “water”, “spring water” etc.).
- In some cases, reported quantities referred to the number of pieces rather than the portion size (e.g., the quantity associated with portion FR1-3, “cooked green banana in coconut cream,” was recorded as six when it should have been 1 – see next image as an example).



2. Convert all quantities into grams

The conversion into grams varied depending on the information collected:

- When the food consumed corresponded to a portion in the food atlas (74% of the cases), the conversion into grams was straightforward, as the weight of each portion was available in the atlas. To obtain the quantity in grams, the reported quantity was multiplied by the corresponding weight in grams from the manual. For instance, if two portions of garlic bread of size 1 (BI1-1) were consumed, the total weight would be calculated as 2 x 57 grams.
- When no portion picture could be identified, three scenarios applied:
 - If a picture of the measurement unit in which the food was consumed was available (for instance, one small glass “M1-4” of water), the quantity was converted into grams using the reported quantity, the volume of one measurement unit, and the density²¹. For instance, the



weight of one cup of coffee corresponding to measurement picture M4-1 with a volume of 380 ml was calculated as: 380 (ml) x 1(g/ml), where 1 g/ml represents the density of coffee.

- If no picture for the measurement unit was available, but the quantity was reported in standard unit such as grams, pounds, or litres, The weight was calculated using the reported quantity and the corresponding density. For instance, if the reported food was “coke”, with the quantity of

²¹ The density corresponds to the weight in grams of one ml of product. It is needed to convert into grams most of the products reported in volumetric unit and for which a density is available (mainly liquid products such as soft drinks, milk etc).

330 ml, the weight was calculated as: $330 \text{ (ml)} \times 1.04 \text{ (g/ml)}$. Where 1.04 g/ml corresponds to the weight in grams of one ml of soft drink (density).

- If no picture was available and the measurement unit was reported in a non-standard unit (e.g., “one pop up ice cream”), an ad hoc weight in grams per unit was provided by SBS, local experts, or extracted from AUSNUT.

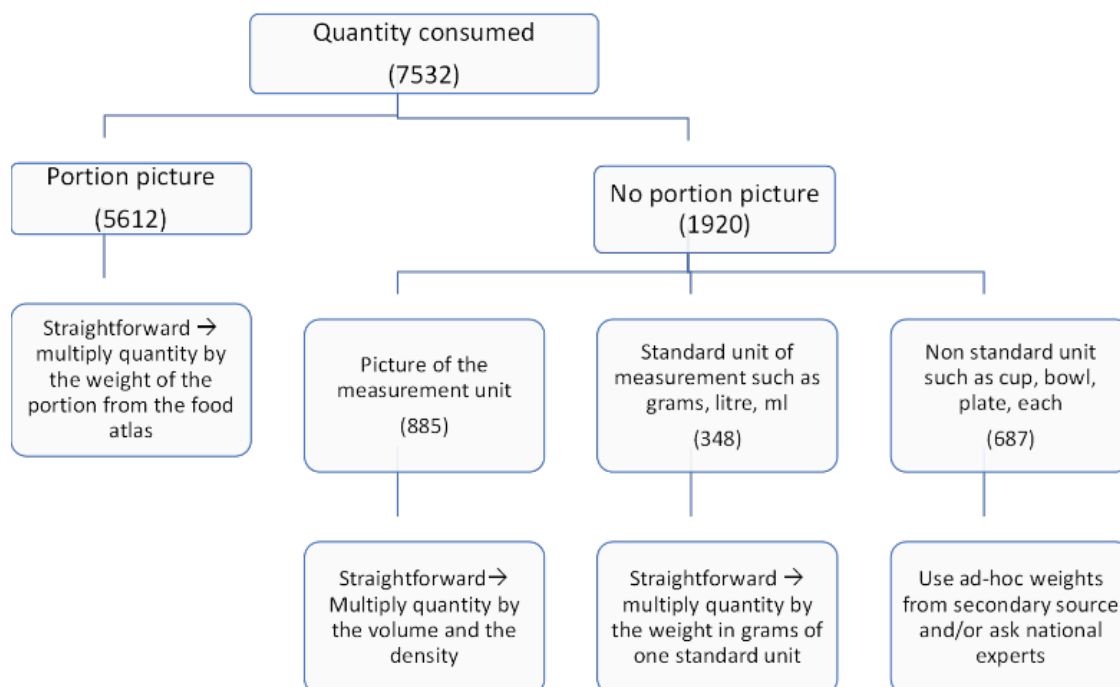


Figure 6. Conversion into grams of the quantities of foods consumed away from home

At the end of this step all the quantities reported were converted into grams

3. Check consistency between the quantity in grams and the corresponding cost

To assess whether the reported cost of meals was consistent with the reported quantity, a “multivariate” approach was adopted²². All the food portions were first recoded into a single food product (for instance, the four portion sizes of “cooked banana”—FR1-1, FR1-2, FR1-3, and FR1-4—were aggregated into a single food product “cooked banana”). The cost of one gram of that food was then estimated. For each food product, we examined the distribution of quantity in grams consumed, the cost reported and the unit price of one gram (for instance, we looked at the distribution of all quantities of “cooked banana” reported by all individuals, the distribution of all costs for “cooked banana” and the corresponding distribution of all price of one gram of “cooked banana”). For each of these distributions, we performed an outlier detection using the Tukey approach based on the inter quartile range (IQR). Through this approach, the intent was to detect whether a reported quantity or cost was an outlier within its own distribution and correct it only if it was not consistent with its respective cost or quantity. This occurred whenever the price per gram was confirmed as an outlier (for instance, the quantity of cooked banana reported by an individual was too high relative to the cost reported, resulting in a price per gram significantly lower than the overall distribution of the price per gram for cooked banana).

At the end of this step, 211 (2.8%) quantities were adjusted using the reported cost and the median price per gram of food product, or by using the median quantity when the reported cost was also an outlier. Additionally, 255 (3.4%) costs were changed.

²² For more detailed information on the “multivariate” approach see the guidance note 6 on food data processing or guidance note 9 on food data cleaning.

4. Convert quantities reported into dietary energy (calories)

To convert the quantities reported from grams to calories, foods for which a portion picture was identified were distinguished from foods for which no portion picture was found.

- Cases for which a portion picture was identified. These cases were straightforward, as the nutrient value was estimated through recipes collected during the cooking session. Nutrient factors were provided per 100 grams of product and were directly applied to the reported quantity (no refuse factor was applied, as nutrient value were provided per 100 grams of edible portion).
- Cases without a portion picture. Out of the 101 food products for which a portion picture was not found in the manual on food portion quantification, 92 corresponded to foods for which a nutrient value could be found in a reference Food Composition table (such as New Zealand CT, AUSNUT, PNDB, or USDA). For the remaining 9 products, the nutrient value was estimated using recipe disaggregation from recipes through web scrapping or SBS.

At the end of this step, a nutrient value was assigned to the final 7,532 food consumption events.

Statistical analysis

The data collected through the FAFH experiment allows for various analysis such as informing FAFH consumption behaviour in Samoa, assessing the gap when using the in-house cost per calorie to estimate the amount of dietary energy consumed away from home, proposing a direct method based on the average amount of dietary energy in a meal consumed away from home, assessing the performance of the 7 day recall module used to collect FAFH in HIES, and analysing the policy implication of the choice of method to estimate dietary energy consumed away from home. This report focuses on the analysis of the gap between the costs per calorie consumed in the house and away from home, and its impact on the prevalence of undernourishment (SDG 2.1.1). All the other analyses will be presented in specific focused studies to be circulated in the course of 2025/2026.

The FAFH experiment was primarily designed to assess the “cost per calorie approach” used to estimate the dietary energy consumed away from home, based on the information collected in HIES. It also aimed to further inform the gap between in-house and away from home costs per calorie. Based on the findings from the experiment, recommendations will be made to update or amend the use of this approach when analysing future HIES conducted in the Pacific.

The results discussed below are based on a final dataset of 7,524²³ observations corresponding to 363 households and 2,483 individuals.

Do we need a multiplier?

To assess whether the cost of one calorie consumed in the house is different from the cost of one calorie consumed AFH, we estimated and compared, for each sampled household, the average costs of one kcal consumed in the house (as collected in Module 2) and AFH (as collected in Module 4).

BOX 4: ESTIMATION OF THE COST PER CALORIE

- Estimation of the “in-house per calorie cost”: Using information from the in-house section filled out by the households selected for the experiment, we first prepared the food data following the process described in guidance note 6²⁴ on food data processing to obtain the calories consumed in the house from each food by each household. For each household, we then estimated the cost of one kcal consumed in the house as total spendings on foods consumed in the house divided by the total

²³ 8 observations corresponding to alcoholic beverages were further removed as alcoholic beverages are reported in the individual module on expenditures in the core questionnaire of the HIES.

²⁴ <https://www.spc.int/digitalibrary/get/vres5>

calories consumed in the house²⁵. We estimated the cost of one kcal consumed in the house, excluding alcohol, as meals consumed away from home in Samoa do not typically include alcoholic beverages.

- Estimation of the “AFH per calorie cost”: For each household, and using information reported in Module 4 (diary), we estimated the total amount spent on foods consumed away from home and we divided by the total calories consumed away from home. This allows us to estimate the AFH per calorie cost as explained in section above.²⁶

The average ratio corresponding to the “AFH per calorie cost” divided by the “in-house per calorie cost” is 2.88, which is significantly different from 1 at 95% confidence level (p-value=0.0000; Table 2). This preliminary finding confirms that the calories from foods prepared and consumed away from home are more expensive than those from foods consumed in the house. On average, people are spending more to consume 1,000 kcal outside of the house compared to the amount they spend to consume 1,000 kcal in their house.

Table 5. Results of T-Test

One-sample t test

Variable	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
multil~r	358	2.882408	.0854343	1.616492	2.71439	3.050426
mean = mean(multilpier)				t =	22.0334	
Ho: mean = 1				degrees of freedom =	357	
Ha: mean < 1			Ha: mean != 1		Ha: mean > 1	
Pr(T < t) = 1.0000			Pr(T > t) = 0.0000		Pr(T > t) = 0.0000	

Further analysis reveals that the ratio differs by population group. This confirms that the kcal cost approach also requires the accounting for socioeconomic and geographic location of the household.

From these findings, it can be concluded that to approximate the “AFH per calorie cost” using the “in-house per calorie cost,” a multiplier higher than 2 is needed in Samoa. Additionally, this multiplier varies based on the characteristics of the household.

Estimation of the DEC_AFH in the Samoa 2023 HIES

The information collected in the AFH experiment was used to estimate the average DEC_AFH in Samoa, using information on in-house food consumption and meals consumed away from home collected in the Samoa 2023 HIES.

²⁵ In HIES, food consumed in the house is collected at the household level and therefore the cost of one kcal consumed in the house is estimated as the total amount spent on foods consumed in the house divided by the total dietary energy consumed in the house disregarding who consumed the food and for which meal event the food was consumed (breakfast, lunch, dinner). However, for specific groups of products such as snacks and non-alcoholic beverages it is possible to estimate the cost of one kcal consumed in the house from these groups if it is confirmed that the products composing these groups are the same than those consumed away from home (for instance a coca cola consumed in the house or in bar). For water consumed away from home, we cannot apply the cost of one kcal as water does not bring energy.

²⁶ The cost of one kcal consumed away from home by meal event can be estimated using different levels of aggregation (i.e. for each meal event one cost per household, or one cost per household/individual or one cost per household/individual/day of consumption). We compared the median cost based on different approaches and we found there was no statistical difference in the mean cost obtained from the various distributions.

For each of the 3,071 households selected for the HIES²⁷, the cost of one kcal consumed in the house was estimated by dividing the total amount spent on food by the total amount of dietary energy consumed in the house. The approach to estimate the cost of one kcal consumed in the house was the same as that described in Box 4 except that:

- The “in-house cost” estimated on all foods consumed in the house was used as a proxy for the cost of one breakfast, lunch or dinner consumed in the house.
- The “in-house cost” of one snack was estimated using only the food products corresponding to snacks and consumed in the house²⁸
- The “in-house cost” for non-alcoholic beverages was also estimated using all the products corresponding to non-alcoholic beverages consumed in the house²⁹

The median of these in-house cost was then estimated for each combination area of residence and quintile of expenditures to account for differences in the cost of one kcal by population group³⁰.

Using data on food consumed away from home collected in the diary from 360 households, the cost of one kcal consumed away from home was estimated for each household. The median of that cost was estimated for each income quintile, area of residence, and meal event (breakfast, lunch, dinner, snack and non-alcoholic beverages).

To follow the recommendations from the GN 6 on food data processing, the sampling weights were used to estimate the median cost of one kcal consumed in the house. However, sampling weights were not used to estimate the median cost of one kcal consumed away from home in the experiment sample. The main reason for this is that the sample from the experiment was too small and applying too much weight to any potential extreme value³¹ could distort the results.

For each combination meal event, expenditure quintile, and area of residence, a multiplier was estimated as the ratio between the cost of one kcal consumed in the house (estimated from the 2023 HIES) and the cost of one kcal consumed AFH (estimated from the AFH diary), resulting in 50 multipliers. See Annex 2 for the multipliers used for the Samoa 2023 HIES.

Using the information on the amount spent on meals consumed away from home collected in the Samoa 2023 HIES, the average DEC_AFH for each meal event (breakfast, lunch, dinner, snack, non-alcoholic beverages)³² was then estimated as follows:

$$DEC_{AFH_{hj}}(kcal) = \frac{Cost_{AFH}(\$)_{hj}}{\underbrace{(Kcal_{cost_{inhouse}}(\$ per kcal) * multiplier_{ji})}_{Kcal_cost_AFH (\$ per kcal)_{ji}}}$$

²⁷ The analysis of the food consumption patterns in Samoa was performed on 3,071 households as a few households presenting implausible food consumption patterns were dropped from the analysis.

²⁸ After confirmation with SBS that the snacks consumed in the house were corresponding to snacks often consumed away from home.

²⁹ After exclusion of koko samoa that is widely consumed in the house but that is not consumed away from home.

³⁰ The median is estimated for each combination area of residence and expenditure quintile as households belonging to these groups are assumed to face very similar dietary energy costs. The dietary energy unit cost is often affected by the household’s place of residence (urban or rural area) and economic status. For example, rich households in urban areas may eat in more expensive restaurants than poor people living in rural areas. The assumption is usually that poor households buy cheap, energy-dense foods, while rich households often buy more expensive (and sometimes healthier) foods.

³¹ For instance, it was observed that in rural area, for the third expenditure quintile 3, the cost of one kcal from dinner consumed AFH was estimated on 3 points only and the weighted median was giving a cost that was three times higher than the non-weighted median leading to a difference in the adjustment factor from 4 to 11.

³² Assuming no difference in the cost of one kcal consumed from breakfast, lunch and dinner as food data collected in HIES are aggregated and do not provide information on in-house consumption by meal event.

Where:

- DEC_AFH_{hj} corresponds to the dietary energy consumed away from home from meal event j consumed by household h
- Cost_AFH_{hj} corresponds to the amount spent on meal j consumed away from home by household j and collected in the 7 day recall module of the Samoa 2023 HIES³³
- Kcal_cost_inhouse_{ji} corresponds to the median cost of one kcal from a representative meal j consumed in the house by expenditure quintile and area of residence i and estimated using information from the Samoa 2023 HIES
- Multiplier_{ji} corresponds to the ratio of the median cost of one kcal consumed away from home from meal event j by expenditure quintile and area of residence i and kcal_cost_inhouse_{ji}.
- Kcal_cost_AFH_{ji} corresponds to the median cost on one kcal consumed away from home of meal event j and consumed by quintile and area of residence

The analysis of the food data collected in the Samoa 2023 HIES, using the information on the cost of one kcal consumed away from home collected through the AFH experiment, finds that, on average, a Samoan consumes around 320 kcal per capita per day away from home. This represents 13% of the total dietary energy consumed and contributes to around 20% of the total amount spent on food. The analysis of the data collected in the diary, however, finds that on average, a Samoan consumes 200 kcal/capita/day³⁴, which suggests a slight underestimation of AFH in the experiment. This underestimation is mainly due to some challenges encountered during the implementation of the experiment, as discussed in Section 5 on lessons learned.

At this stage, the gap in the dietary energy consumed, as estimated using both direct (information collected from Module 4) and indirect approaches (information collected from Module 3 and combined with Module 2), does not allow us to conclude that the indirect approach accurately approximates the true amount of dietary energy consumed away from home at the individual level. The fact that some consumption might not be fully captured in the diary (such as consumption during To'onai) and that such consumption might be dense energy and less expensive than some of the consumption reported in the diary could lead to a slight overestimation of the “true” cost per calorie consumed away from home.

Implications for estimates of key indicators

A value of 1.25 for the multiplier, as currently used when analysing food data in HIES conducted in PICTs after 2018, would have brought the average amount of dietary energy consumed away from home in Samoa to 530 kcal per capita per day. Therefore, the choice of the multiplier when estimating DEC_AFH using the “in-house kcal cost approach” is instrumental when preparing food data collected in HIES for poverty or food security analysis.

Using information collected in the Samoa 2023 HIES, a short sensitivity analysis was conducted to assess the impact of the choice of the multiplier on SDG 2.1.1 the prevalence of undernourishment (PoU). The prevalence of undernourishment corresponds to the percentage of the population whose dietary energy intake is lower than the amount of dietary energy needed to be in good health and maintain an active life. It has been used by FAO to monitor and report on global hunger since 2000³⁵.

The prevalence of undernourishment is estimated using a parametric approach based on three parameters, under the assumption that the dietary energy consumption follows a log normal distribution. The three parameters

³³ Note that the 7-day recall module on food away from home included in HIES is administered at individual level but the information collected is then brought to the household and the total amount of dietary energy consumed away in the household is divided by the size of the household to obtain the per day per capita dietary energy consumed away from home.

³⁴ Considering all individuals, those who consumed away from home and those who do not to obtain a representative average of dietary energy consumed away from home per capita per day.

³⁵ First to report on progress toward the Millennium Development Goals (MDGs), and then to report on SDG 2.1.1 after 2015

needed to estimate the prevalence of undernourishment are: the average amount of dietary energy consumed in a country (DEC), the coefficient of variation³⁶ (CV)— an indicator of dispersion of dietary energy consumption within the population, and the minimum dietary energy requirement (MDER), which is the threshold below which people are considered undernourished. See Annex 3 for more details on the methodology.

The sensitivity analysis was conducted using the food data collected in the Samoa 2023 HIES, where we estimated the average DEC using different multipliers to account for the cost differential between in-house and away from home consumption. We then analysed the impact on PoU through changes in DEC and the CV, assuming no change in the minimum dietary energy requirement (MDER) which was kept constant at 1,843 kcal/capita/day³⁷.

We assumed different values for the multiplier:

- Scenario 1 – A multiplier of 1 (no difference between the cost of one kcal consumed in the house and one kcal consumed AFH) was applied to the median in-house kcal cost estimated for each combination of expenditure quintile and area of residence. This is the approach currently used by FAO when estimating DEC_AFH.
- Scenario 2 – A multiplier of 1.25 applied on median in-house kcal cost, estimated for each combination of expenditure quintile and area of residence. This is the same approach currently used by SPC to estimate DEC_AFH from recent HIES conducted in the PICTs.
- Scenario 3 – a multiplier of 1.25 applied on a “flat” median in-house kcal cost. This is the approach used by the World Bank to estimate DEC_AFH³⁸.
- AFH experiment – The multiplier, calibrated on the cost of one kcal consumed away from home, estimated from the Samoa AFH experiment, was applied to the median in-house kcal cost estimated for each combination of expenditure quintile and area of residence. This corresponds to the current estimate of DEC_AFH used for the analysis of food consumption patterns in Samoa and the estimation of SDG 2.1.1

Table 6. Impact on SDG 2.1.1 of the decisions made to estimate DEC_AFH

	Scenario 1 “FAO”	Scenario 2 “SPC”	Scenario 3 “WB”	AFH experiment
Average AFH DEC (kcal/capita/day)	651	531	608	320
Average DEC (kcal/capita/day)	2,977	2,857	2,934	2,646
CV (%)	22	22	24	22
MDER (kcal/capita/day)	1,843	1,843	1,843	1,843
PoU (%)	2	3	3	6
Number of people undernourished	4,275	6,464	7,269	12,637

Table 6 clearly shows that the decisions made when estimating DEC_AFH can impact on the estimation of key indicators and the message conveyed to policy makers. The most important and impactful decision lies in the choice of the multiplier.

This sensitivity analysis was performed using data from Samoa, where the cost of eating outside the house may be more expensive than in other PICTs. The multiplier used may not reflect the differential in other countries, as it was

³⁶ Obtained as the standard deviation divided by the mean of the distribution of DEC.

³⁷ Estimated using normative requirements from the report of the joint FAO/WHO/UNU Expert Consultation in Rome 2001 (<https://www.fao.org/4/y5686e/y5686e00.htm>) and the information on the age and gender distribution of the population in Samoa and the median height per class of age and gender collected in the HIES conducted in Tonga in 2015.

³⁸ With the difference that WB estimates this cost on the first quintile of the population only while we estimated it using all households.

1.25 in RMI and 1.5 in Papua New Guinea³⁹, based on analysis of spendings to eat outside and inside the house. The major difference in the multiplier tends to confirm the need for additional research work, eventually, the implementation of similar protocols in other countries to obtain a multiplier that would be more representative by subregions of the Pacific.

4. Conclusion

Through the food data collected in HIES, it is possible to derive the distribution of the dietary energy consumed in a country. This information represents an important component of poverty analysis and is central to the measurement of hunger through the prevalence of undernourishment indicator.

The amount of dietary energy consumed in a country is coming from two main types of foods: foods consumed in the house (in-house foods) and foods prepared and consumed away from home (away from home foods). While the estimation of dietary energy from in-house food consumption collected in HIES is direct and straightforward (following international standards), estimating dietary energy from foods prepared and consumed away from home using information collected in HIES is more challenging. It relies on an indirect approach based on the cost per calorie from in-house consumption and two main assumptions. The first assumption is that the amount spent on foods prepared and consumed away from home is accurately reported. The second assumption is that the food basket for foods consumed away from home is very similar to the basket of foods consumed in the house. While much research has been conducted to improve the information on amount spent on foods consumed away from home, little research has been done on the composition of away from home consumption. If the second assumption does not hold true, it has never been formally demonstrated using direct measurement.

Through this experiment conducted in Samoa by SBS in collaboration with SPC, we proposed a protocol to collect data on foods prepared and consumed away from home using an innovative approach to directly measure dietary energy consumed away from home. This experiment demonstrated that one calorie consumed away from home is far more expensive than one calorie consumed in the house. Decisions made when estimating dietary energy consumed away from home using information collected in HIES can have significant policy implications. One key decision lies in determining the value of the multiplier needed to adjust the cost per calorie estimated on in house food basket to proxy the cost per calorie from foods prepared and consumed away from home. This experiment found that a multiplier of 1.25, estimated using data collected in the 2018 RMI experiment, would be too low for a country like Samoa, where the multiplier was found to be higher than 2.

Based on all information available, it is difficult to formally recommend a specific multiplier at this stage. While the experiment was successfully implemented in Samoa, some challenges were encountered that might lead to a slight overestimation of the multiplier. One of the main challenges was that most of the consumption occurring on Sundays was not properly captured. This consumption mostly consists of potluck meals, where people bring food to share, which may be dense in energy and less expensive than many foods reported in the diary or acquired from formal food vendors, such as in a restaurant. If this consumption were properly captured, the cost of one calorie consumed away from home would likely be slightly lower than that actually reported in the experiment, reducing de facto the value of the multiplier. Therefore, it is recommended to continue research in this field and in the meanwhile use a multiplier close to 2 when estimating dietary energy consumed away from home using HIES conducted in Pacific countries.

To make the optimal use of the information carried in this experiment, various studies focusing on specific research area will be published and disseminated in the course of 2025/2026:

- A study focusing on the away from home consumption behaviour in Samoa – in this study we will try to address questions such as who eats what, main places of consumption, main foods consumed, amount spent, etc;

³⁹ Even if these two values were estimated using spendings and not a direct measurement of dietary energy as from this experiment.

- A study to assess the performance of the 7-day recall module on meals consumed away from home – in this study we will compare the individual module incorporated in the HIES to collect meals away from home (based on the WB/FAO guidelines) with the information collected in the diary – keeping in mind the challenges faced when implementing the experiment and the potential light underestimation of the “true” away from home consumption;
- A study to compare the calorie per meal approach with the cost per meal approach; and
- A study to assess the impact of the adjustment factor on poverty analysis and evaluation how the choice of the multiplier affects the food basket and poverty estimate.

Finally, this experiment through its innovative and complex nature, has been rich in lessons for SPC, which was exposed to a new way of collecting information using visual aids, and for SBS, which faced additional challenges due to the adoption of the harmonized HIES (switching from diary to recall, PAPI to CAPI, etc). Annex 4 provides a summary of some of the lessons learned during the implementation stages of the experiment, highlighting both things that worked well and things that could be improved in the future implementations of such experiments in other countries.

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Annex 1. Module 4 – Diary

HOW TO FILL IN THIS DIARY									
		Mon	Tue	Wed	Thu	Fri	Sat	Sun	(dd/mm/yy)
Day 03		☐	☐	☐	☐	☐	☐	☒	Date 27 11 22
Name of the respondent: <u>Tupae</u>								Food or beverage 1	
Q1 Did you consume some food or beverage outside of your house today? <i>Food and/or beverage refer only to food and/or beverages both prepared and consumed outside of your house.</i>								☒ Yes ► Q2 ☐ No ► Next day	
Q2 Did you consume the food or beverage during:								☐ Breakfast? ☐ Lunch? ☒ Dinner? ☐ For snack?	
Q3 What is the name of the food or beverage you consumed today outside of your house? <i>Complete each column before moving to the next food or beverage</i>								Enter name: <u>Garlic bread</u> ► Q7 ☐ Don't know ► Q4	
Q4 Which picture from the food atlas best describes the food you consumed today?								Enter code: _____ ► Q7 ☐ Not in the atlas ► Q5	
Q5 What were the main ingredients contained in the food or beverage consumed? <i>List all the ingredients of your food or beverage, starting from the main ones. For example: if you consumed "fish in taro leaves and coconut cream" then write: fish, taro leaves and coconut cream.</i>								1. _____ 2. _____ 3. _____ 4. _____	
Q6 What was the cooking method of the food? <i>For example, if the food was mainly fried, boiled, baked, cooked using UMU method, etc. Do not write anything if not applicable (for instance raw foods such as sashimi, ice creams, etc.).</i>								1. _____ 2. _____ 3. _____ 4. _____	
Q7 Which picture from the food atlas best describes the PORTION of the food you consumed? <i>For example if you were served a plate of fried canned corned beef (falai pisupo) but ate only half of the plate enter the code corresponding to the picture of the half plate (code F14-Q2 in the food atlas).</i>								Enter code: <u>B11 - 2</u> ► Q10 ☐ Not in the atlas ► Q8	
Q8 Which picture from the food atlas best describes the MEASUREMENT UNIT in which you want to report the quantity consumed? <i>For example, if you had one small glass of wine, enter the picture corresponding to a small glass (code M2-4, or M2-5 or M2-6 in the food atlas); if you had 3 big spoons of coco rice, enter the picture corresponding to the spoon more similar to the one you used to serve the food (for instance code M5-1 in the food atlas).</i>								Enter code: _____ ► Q10 ☐ Not in the atlas ► Q9	
Q9 In which unit of measurement would you like to report the quantity? 1. Household measurement (cup, bowl, tin, whole glass...) 3. Standard unit (gram, litre, etc.) 2. Standard portion (slice, pack, stick...) 4. Other								Enter unit of measurement: _____	
Q10 What was the quantity you consumed? <i>For portions: number of portions. For household measurement: Number of servings. For standard or other units: quantity in units. If the portion identified in Q6 corresponds to the quantity you consumed, then just enter 1.</i>								Enter quantity: <u>1</u>	
Q11 Where did you procure the food and beverages? <i>Enter the place where you procured the food and beverages, not the one where you consumed it. For example, if you procured the food on a street stand and consumed it in the office, then write "street stand". In case of restaurants, café, bar, night club or "Other" please enter the name of the establishment (optional).</i>								Enter place reference: <u>1</u> Name of establishment (optional): _____	
Place references: 1. Restaurant/café 4. School canteen 7. Church 2. Bar/night club 5. Work canteen 8. Other 3. Street stand/Market 6. Friends or relatives									
Q12 Was the meal: 1. Purchased 2. Received for free 3. Received in-kind.								Enter code reference: <u>1</u>	
Q13 How much did you pay for the food or beverage? In case you did not pay for it, how much would you have paid to buy it? <i>If you consumed for free from a buffet, enter only once the total amount in Tala that you would have paid for all the food you consumed.</i>								Enter value: <u>8</u>	

Annex 2. Multipliers applied to the Samoa 2023 HIES AFH data

Meal event	Area of residence	Expenditure quintile	Median cost of one kcal consumed AFH (SAT/kcal)	Median cost of one kcal consumed in the house (SAT/kcal)	Multiplier
Breakfast away from home	Urban	1	0.0057	0.0025	2.27
Breakfast away from home	Urban	2	0.0056	0.0029	1.93
Breakfast away from home	Urban	3	0.0040	0.0033	1.22
Breakfast away from home	Urban	4	0.0051	0.0036	1.42
Breakfast away from home	Urban	5	0.0064	0.0045	1.44
Breakfast away from home	Rural	1	0.0040	0.0025	1.59
Breakfast away from home	Rural	2	0.0055	0.0026	2.09
Breakfast away from home	Rural	3	0.0045	0.0028	1.63
Breakfast away from home	Rural	4	0.0057	0.0031	1.86
Breakfast away from home	Rural	5	0.0062	0.0038	1.62
Lunch away from home	Urban	1	0.0064	0.0025	2.53
Lunch away from home	Urban	2	0.0050	0.0029	1.72
Lunch away from home	Urban	3	0.0061	0.0033	1.84
Lunch away from home	Urban	4	0.0085	0.0036	2.38
Lunch away from home	Urban	5	0.0099	0.0045	2.21
Lunch away from home	Rural	1	0.0060	0.0025	2.39
Lunch away from home	Rural	2	0.0065	0.0026	2.45
Lunch away from home	Rural	3	0.0065	0.0028	2.37
Lunch away from home	Rural	4	0.0070	0.0031	2.28
Lunch away from home	Rural	5	0.0083	0.0038	2.17
Diner away from home	Urban	1	0.0093	0.0025	3.66
Diner away from home	Urban	2	0.0065	0.0029	2.24
Diner away from home	Urban	3	0.0068	0.0033	2.05
Diner away from home	Urban	4	0.0089	0.0036	2.47
Diner away from home	Urban	5	0.0149	0.0045	3.31
Diner away from home	Rural	1	0.0061	0.0025	2.44
Diner away from home	Rural	2	0.0107	0.0026	4.08
Diner away from home	Rural	3	0.0121	0.0028	4.38
Diner away from home	Rural	4	0.0084	0.0031	2.72
Diner away from home	Rural	5	0.0133	0.0038	3.47
Non alcoholic drinks away from home	Urban	1	0.0167	0.0357	0.47
Non alcoholic drinks away from home	Urban	2	0.0221	0.0355	0.62
Non alcoholic drinks away from home	Urban	3	0.0234	0.0312	0.75
Non alcoholic drinks away from home	Urban	4	0.0202	0.0317	0.64
Non alcoholic drinks away from home	Urban	5	0.0179	0.0324	0.55
Non alcoholic drinks away from home	Rural	1	0.0228	0.0404	0.56
Non alcoholic drinks away from home	Rural	2	0.0228	0.0347	0.66
Non alcoholic drinks away from home	Rural	3	0.0176	0.0347	0.51
Non alcoholic drinks away from home	Rural	4	0.0221	0.0348	0.63
Non alcoholic drinks away from home	Rural	5	0.0163	0.0347	0.47
Snacks away from home	Urban	1	0.0080	0.0063	1.27
Snacks away from home	Urban	2	0.0101	0.0050	2.03
Snacks away from home	Urban	3	0.0042	0.0045	0.94
Snacks away from home	Urban	4	0.0094	0.0049	1.91
Snacks away from home	Urban	5	0.0066	0.0042	1.56
Snacks away from home	Rural	1	0.0044	0.0049	0.88
Snacks away from home	Rural	2	0.0038	0.0052	0.72
Snacks away from home	Rural	3	0.0045	0.0045	0.99
Snacks away from home	Rural	4	0.0056	0.0045	1.25
Snacks away from home	Rural	5	0.0069	0.0048	1.43

Annex 3. Methodological Annex – the prevalence of undernourishment

Definition: Undernourishment is defined as the condition of an individual whose habitual food consumption is insufficient to provide, on average, the amount of dietary energy required to maintain a normal, active, healthy life.

How it is reported: The SDG2.1.1 indicator is reported as a prevalence and is denominated as “prevalence of undernourishment” (PoU), which is an estimate of the percentage of individuals in the total population that are in a condition of undernourishment.

Methodology: To compute an estimate of the prevalence of undernourishment in a population, the probability distribution of habitual dietary energy intake levels (expressed in kcal per person per day) for the average individual is modelled as a parametric probability density function (pdf), $f(x)$. The indicator is obtained as the cumulative probability that the habitual dietary energy intake (x) is below the minimum dietary energy requirements (MDER) (i.e., the lowest limit of the range of energy requirements for the population’s representative average individual) as in the formula below:

$$PoU = \int_{x < MDER} f(x|\theta) dx$$

where θ is a vector of parameters that characterizes the pdf. The distribution is assumed to be lognormal, and thus fully characterized by only two parameters: the mean dietary energy consumption (DEC), and its coefficient of variation (CV).

Challenges and limitations: While formally the state of being undernourished or not is a condition that applies to individuals, given the data usually available on a large scale it is impossible to reliably identify which individuals, in a certain group are actually undernourished. Through the statistical model described above, the indicator can only be computed with reference to a population or a group of individuals for which a representative sample is available.

Because of the under estimation of dietary energy in urban areas due to under reporting of food consumed away from home in the 2018 HIES, the prevalence of undernourishment for Samoa cannot be further disaggregated.

Due to the probabilistic nature of the inference and the margins of uncertainty associated with estimates of each of the parameters in the model, the precision of the PoU estimates is generally low. While it is not possible to formally compute margins of error around PoU estimates, these are expected to likely exceed 5 percent in most cases. For this reason, in case of Samoa we report a prevalence of undernourishment higher than 2.5% and lower than 5%.

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Annex 4. Challenges and lessons learned

Activity	What worked well	Challenges	How to improve
Planning and design			
Experiment design	Expert group was productive and low cost	Time delays from PACSTAT innovative experiment proposal to approval, then delays in grant approval, then delays in disbursement, then delays in reimbursement	Reduce the administrative inefficiencies of PACSTAT granting
Grant operations	PACSTAT research proposal was effective to support experiment design and granting	Grant delays forced SBS to pay all upfront costs (Phase 1); most fieldwork costs committed to be paid via the grant were not paid as at Sept 2024 The Grant was approved via both the PACSTAT procedures, and by SPC's Grants Management Unit resulting in long delays and heavy administration	PACSTAT to complete grant disbursement, and improve efficiency of granting processes Use SPC or World Bank granting procedures, but not both
Meal list – web search	Very effective means to collect list of meals (incl. prices and some recipes), location of vendors (incl. GPS coordinates)	Bias towards urban and formal business	Fill data gaps by completing physical visits to food vendors
Meal list – use of 2018 HIES	It was an accessible data source that provided some meals to be listed	Only prepared foods for in-house consumption were reported in the 2018 HIES, not information on FAFH	Ensure survey microdata are curated and accessible

Activity	What worked well	Challenges	How to improve
Meal list - crowd sourcing	Very effective and cost-effective means to collect a small amount of demographic, geographic, meal name, vendor name, and photo data Instrument developed to collect data based on both a single meal consumed by an individual, or a buffet (or To'onai, the important Samoan Sunday lunch) where multiple meals are shared SurveySolutions with a tiny URL and a QR code was highly efficient	Funding delays constrained this activity to just 2-weeks with little testing, promotion, or opportunity for multiple rounds With more time, there were options to have more targeted campaigns, such as one just for To'onai, or specific campaigns for populations, such as rural and urban Age and area bias probably to youth and urban, but smartphone and Facebook use is highly prevalent in Samoa thereby not being an excluding factor for those populations	Continue to explore opportunities to crowdsource data
Meal list - focus group discussion	Very effective means to access information on sources of FAFH that are important, yet not from a formal establishment or a To'onai, such as school, catered events, and Church	Funding delays constrained preparation for this event which hindered the objective. With more time, the event itself may have been more productive, but there would have been more opportunity for follow up Interesting groups identified with lots of knowledge, but limited time for follow up Kids at colleges and primary schools were not included, so we missed what they consume AFH.	More time on such activities, but continue to engage with food networks in Samoa (perhaps for this project dissemination) Ensure all population groups are represented and participate to such meeting
Meal list - physical visits	Very effective means to collect foods available at informal establishments, and often to view how the food is prepared Accessible coverage of informal commercial food vendors in small island economies	With better preparation the food could have been purchased, de-cooked, and data measured on the spot; this would provide opportunity to interview the vendor about the ingredients Price data could have been collected	This sort of collections could be included in regular CPI collections (including the collection of unit values, which are useful for food statistics)

Activity	What worked well	Challenges	How to improve
Kitchen, Chef, Photographer and nutritionists procurement	The chef, restaurant, photographer, and nutritionists, who were procured to the project were all of excellent quality, and they were necessary, and they added value Having the nutritionist procured from the design phase adds enormous value, and it's essential this person is there during the collection of nutrient data Having two additional nutritionists from SPC available to travel last minute was also instrumental to the success of the cooking and shooting session	Grant disbursement delays forced reduced time to procure and get value for money More cost-effective options whereby a kitchen is hired (not commercial, perhaps an AirBNB) and local retired people are procured to prepare local foods using local cooking methods	More time allocated to procurement More creativity in planning who, how, and where, to cook foods - some commercially cooked foods are necessary, but many Samoan foods and prepared food available for sale by informal vendors do not require a commercial chef or kitchen and cost savings can therefore be achieved
Engagement with SPC in pre-fieldwork activities (both SDD and PHD)	Aside from the grant disbursement issues, SBS and SPC collaboration on the experiment and HIES has been positive with SBS adopting new methods and instruments for the HIES, and the FAFH experiment is new for everyone Collaboration continuing with the food data processing and HIES reporting underway The inclusion of nutritionists from SPC as collaborators with SDD on the experiment was highly positive and a contributing success factor SBS successfully mobilised finance and people to ensure this phase of the experiment was successfully executed	Granting issues identified above put a lot of pressure on SBS and SPC which complicated the collaboration Everything was new - new collaboration, new donor, new HIES method and mode, complex experiment, project personnel scattered around the world	Continue engagement throughout all HIES and experiment activities

Activity	What worked well	Challenges	How to improve
Pre-testing	SBS included pre-testing in the HIES, which was completed, but not for the FAFH experiment	Time delays meant that the experiment instrument was not available until just before field work	Address disbursement issues
HIES training, presentations, and manuals	SBS staff quickly understood HIES concepts, and actively participated in training and development of the manual resulting in efficient capacity transfer which improved the HIES and experiment significantly The new HIES method meant there were fewer trainees and a more efficient personalised training	The pre-test training became the main HIES training, while the training before HIES field work was more a refresher for the HIES training and a training for the FAFH experiment HIES manual provided late and certain sections obsolete because the questionnaire changed	Standardise HIES instrument and therefore training modules SPC to develop a HIES manual Do not change the questionnaire during field work and spend more time on testing and review of the data prior to field work
FAFH experiment training	SPC was able to be in Samoa during the HIES and FAFH experiment training, and during the first weeks of field operations SBS had been active with HIES pre-testing and all of SBS and the enumeration team were well versed on the questionnaire and CAPI system Adding experiments to surveys is efficient	FAFH diary and interaction with the food quantification manual was complicated	Finalise the instrument earlier so it can be tested, and training manuals and resources can be properly developed and delivered
HIES method (maps, recall, all year)	HIES recall method with fewer field staff, a rolling 12-month sample, one-visit, CAPI, was a first for Samoa and it was embraced and has thus far appeared a success	Maps were initially an issue due to some divergence between SBS and SPC produced maps, and difficulty in accessing certain dwellings Field staff retention is an issue and recruitment and training was necessary throughout the 12-month HIES	SBS engaging with SPC on development of geospatial and sampling statistics

Activity	What worked well	Challenges	How to improve
HIES instrument (Survey Solutions)	Survey Solutions was well received and adopted by SBS Questionnaire design was efficiently done in partnership between SBS and SPC No apparent issues with internet accessibility and data transfer SBS was extremely active in acting as Supervisors/HQ for the HIES and they reviewed the data comprehensively and communicated a lot with field teams	Questionnaire continued to change, even after field work SBS did not know how to use the Survey Solutions dashboard to monitor HIES field operations (fortunately, through their collaboration with ADB, they received support and were able to utilise this function throughout the HIES)	Finalise the instrument earlier so it can be tested and training manuals and resources can be properly developed and delivered
FAFH instrument - food quantification and diary	The food quantification manual and diary interacted well and provided numerous options for quantifying food, including by direct match of food and quantity, match of food with similar composition and quantity, and use of the provided units or other respondent-selected quantities and units. Receipts kept by households were very useful in completing the diaries	The diary was complicated to complete, and this often resulted in a quasi-recall survey where the SBS field staff completed the diary during their three visits	SBS provided good instruction to the respondents to make a list of their FAFH consumption so it could more easily be recalled and transcribed to the diary Encourage individuals to keep the receipts when purchasing and consuming away from home.
Engagement with SPC during fieldwork	The SBS and SPC team have very open communication channels and dialogue on technical matters is frequent and unencumbered by distance	There's limited feedback from SPC on data quality during field operations	The provision of advice by ADB on use of the Survey Solutions functionality for field work monitoring was important and valuable input SPC to standardise the instrument and to develop programs to monitor HIES field operations

Data processing and analysis

Activity	What worked well	Challenges	How to improve
Data coding and organising	Done by SPC using what's becoming fairly standard code Great engagement with SBS on coding (ISIC, ISCO, COICOP), translation	It's quite challenging to collaborate on this process as it's often manual labour of data structuring (wide to long, or vice versa), code matching and labelling, variable naming, and consistency checks	SPC to start to use GitHub to share code (and data under controlled environments)
Preparation of the income and expenditure aggregates	SBS and SPC have reviewed income and expenditure aggregates and identified errors and areas to further investigate improving confidence in the data	The current HIES method (using recall for food consumption, and CAPI) is quite different to the 2018 method so comparability is a challenge	
Food data processing	SBS provided great support to SPC in standardising food consumption estimates SPC workshop on food data processing engaged SBS in the process and was an opportunity to review the and improve the data, and to give SBS Stata training	SBS has limited exposure to food data processing, so the concepts were all new SBS need more opportunity to use HIES data and Stata	Continue capacity transfer to SBS
FAFH experiment	SBS did a great job in collecting the data for the FAFH experiment	Coding of the information and preparation of the data for analysis was challenging due to the type and nature of the data collected	Improve material for training of enumerators and make sure coding instructions are clear
HIES reporting	Ongoing activity	Ongoing activity	Ongoing activity
HIES data use (GDP, CPI, poverty, food)	Engagement with IMF has thus far been productive	Ongoing activity	Ongoing activity
Engagement with SPC in post fieldwork activities	Ongoing activity	Ongoing activity	Ongoing activity